1. Drugs

Drugs: Any substance taken into the body that modifies or affects chemical reactions in the body. All drug metabolism is done in the **Liver.**

1.1. Antibiotics

- Antibiotics work by disrupting crucial systems that bacteria need to survive and multiply, such as their cell walls, DNA, or protein synthesis, while not harming human cells.
- Some bacteria are resistant to antibiotics, which reduces the effectiveness of antibiotics.
- The development of resistant bacteria such as MRSA can be minimized by **limiting antibiotics only when essential and ensuring treatment is completed.**
- Antibiotics don't work on viruses because they **do not have a cell wall** and make the host cell perform their tasks.

1.2. Antibiotic Resistance

Antibiotic-resistant bacteria can be reproduced through the theory of **natural selection**:

- Mutation, giving rise to variation.
- Antibiotics kill bacteria without changing genes
- Competition for food space, among others
- Reproduce via binary fission
- Then, alleles are passed on to offspring to reproduce.

An example of antibiotic-resistant bacteria is **Methicillin-resistant Staphylococcus aureus (MRSA)**

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